1. Interpreting Mathematical Results and Recontextualizing Them in Real-World Scenarios

(Corresponds to Sections 1, 5, and 9 of "interpret.pdf")

- Question Content: The task requires students to compare the payment schemes of two newspapers (Zedland Star and Zedland Daily) and select the correct graph representing their compensation structures.
- Relevance Analysis:
  - Students must understand how mathematical representations (e.g., linear functions, piecewise functions) reflect real-world salary structures.
  - They need to connect abstract mathematical graphs to concrete practical scenarios (newspaper sales earnings) to determine which graph accurately depicts the two payment schemes.
  - This directly aligns with "interpret.pdf" principles such as "interpreting mathematical results back into real-world contexts" and "understanding the relationship between problem situations and mathematical representations of solutions."

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2. Evaluating the Reasonableness of Mathematical Conclusions Within Given Contexts

(Corresponds to Sections 2 and 4 of "interpret.pdf")

- Question Content: Students must identify which of four graphs correctly represents the newspapers' payment schemes.
- Relevance Analysis:
  - Students must consider real-world factors: Zedland Daily offers a fixed weekly wage (60 zeds/week) plus sales bonuses, while Zedland Star uses tiered pricing (0.20 zeds for the first 240 copies, then 0.40 zeds thereafter).
  - If a graph shows Zedland Daily's income as zero when sales are zero, it contradicts reality (due to the fixed wage) and can be eliminated.
  - This embodies "interpret.pdf" guidelines like "assessing whether mathematical results or conclusions are reasonable within the problem context" and "using mathematical tools to evaluate solution validity."

3. Recognizing Limitations of Applied Models

(Corresponds to Sections 3, 6, and 8 of "interpret.pdf")

- Question Content: While the question does not explicitly ask about model limitations, it implicitly involves the approximation of reality through mathematical models (linear/piecewise functions).
- Relevance Analysis:
  - In reality, newspaper earnings could be influenced by additional factors (e.g., weather, location, demand fluctuations), but the problem simplifies this to sales volume and fixed formulas.
  - Students who think critically might recognize that this simplified model overlooks real-world complexities.
  - This aligns with "interpret.pdf" points such as "identifying limitations of models used to solve problems" and "acknowledging the scope and constraints of mathematical concepts and solutions."

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4. Constructing and Communicating Context-Based Explanations and Arguments

(Corresponds to Section 7 of "interpret.pdf")

- Question Content: Although the question only requires selecting an answer, if extended to an open-ended format, students would need to justify their choice.
- Relevance Analysis:
  - Students would combine mathematical reasoning with real-world context to explain their selection (e.g., "Zedland Daily earns 60 zeds even with zero sales, so its graph must have a y-intercept").
  - This matches "interpret.pdf" guidance on "constructing and communicating explanations and arguments grounded in problem contexts."

1. 解释数学结果并将其还原到现实情境（对应《interpret.pdf》第1、5、9条）

- 题目内容：该题目要求学生比较两家报纸（Zedland Star 和 Zedland Daily）的薪酬方案，并选择正确的图表来表示它们的支付方式。
- 相关性分析：
  - 学生需要理解数学表达（如线性函数、分段函数）如何反映现实世界中的薪酬结构。
  - 需要将抽象的数学图表与具体的实际问题（卖报收入）联系起来，判断哪个图表正确描述了两种薪酬方案。
  - 这直接对应《interpret.pdf》中提到的“将数学结果解释回现实世界的情境”以及“理解问题情境与数学解表示之间的关系”。

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2. 判断数学结论在给定情境下是否合理（对应《interpret.pdf》第2、4条）

- 题目内容：学生需要判断四个选项中哪个图表准确反映了两种报纸的薪酬支付方式。
- 相关性分析：
  - 学生必须考虑现实因素，比如 Zedland Daily 提供了固定工资（60 zeds/周）+ 额外销售奖励，而 Zedland Star 采用阶梯式计价（前240份0.20 zeds，之后0.40 zeds）。
  - 如果某个选项的图表显示 Zedland Daily 的收入在销量为 0 时也为 0，则显然不符合现实（因为有固定工资），因此可以排除。
  - 这体现了《interpret.pdf》中提到的“判断数学结果或结论在问题情境下是否合理”以及“使用数学工具评估解决方案的合理性”。

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3. 识别所用模型的局限性（对应《interpret.pdf》第3、6、8条）

- 题目内容：虽然题目本身没有明确要求讨论模型的局限性，但隐含地涉及了数学模型（线性/分段函数）对现实世界的近似。
- 相关性分析：
  - 现实中，卖报收入可能受更多因素影响（如天气、地段、报纸需求波动等），但题目仅考虑销量和固定公式。
  - 学生如果深入思考，可能会意识到这种简化模型忽略了现实复杂性。
  - 这对应《interpret.pdf》中的“识别用于解决问题的模型的局限性”以及“认识数学概念和解法的范围和限制”。

4. 构建并传达基于问题情境的解释和论证（对应《interpret.pdf》第7条）

- 题目内容：虽然题目只要求选择正确答案，但若扩展为开放性问题，学生需要解释为何某个图表正确。
- 相关性分析：
  - 学生需结合数学推理和现实情境，说明选择依据（例如：“Zedland Daily 在销量为 0 时仍有 60 zeds 收入，因此图表应在 y 轴上有截距”）。
  - 这符合《interpret.pdf》中“构建并传达基于问题情境的解释和论证”的要求。